

CS m146 HW#3

1. $J(\mu, z) = \sum_{n=1}^N \|x_n - \mu_{z_n}\|^2$ $x_n \in \mathbb{R}^d$, $z_n \in \{1, \dots, k\}$ cluster, $\mu_c \in \mathbb{R}^d$ is a centroid K means

Lloyd's 1. Assignment - Centroids fixed 2. update each z_n to minimize J . (change points, clusters)

2. Update - Assignments Fixed, update each centroid μ_c to minimize J .

a. Show that the value that minimizes J is the mean of the points assigned to C .

$$\mu_c^* = \frac{1}{|C_c|} \sum_{n: z_n=c} x_n \quad ; \quad C_c = \{n: z_n=c\} \quad , \quad J = \sum \|x_n - \mu_c\|^2 + \text{terms not involving } \mu_c$$

$$\nabla \mu_c = -2 \sum_{n: z_n=c} (x_n - \mu_c) = 0 \quad \sum_{n: z_n=c} x_n = \sum_{n: z_n=c} \mu_c \quad \text{all labels } n \text{ s.t. point } n \text{ is assigned to cluster } C$$

$$\sum_{n: z_n=c} x_n = |C_c| \mu_c \quad \text{so} \quad \mu_c^* = \frac{1}{|C_c|} \sum_{n: z_n=c} x_n$$

b. Fixed centroids $\rightarrow$ best cluster for each x_n $J = \sum \|x_n - \mu_{z_n}\|^2 = \|x_1 - \mu_{z_1}\|^2 + \|x_2 - \mu_{z_2}\|^2 + \dots$ only depends on a constant

$$\text{so for point } x_n \text{ just pick the center that's closest: } z_n^* = \underset{c \in \{1, \dots, k\}}{\text{argmin}} \|x_n - \mu_c\|^2$$

c. I. monotonically decreases II. Converges. J decreases after assignment step because if any point gets reassigned to a closer centroid J strictly decreases. For update step assignments are fixed and move each centroid to mean of its cluster so optimal centroid so J must decrease

so $J^{t+1} \leq J^t$. There are only a finite # of ways to assign n points to k clusters so we can't get stuck.

never revisits same one because J would have to be equal but J is already lower. so Alg must eventually converge.

d. why J converges to local min? Lloyd's 1985 is a non-convex func so it descends on one of many minimizers none of which are guaranteed to be the global min. This is why randomizing the starting centroids position can lead to different results often better ones than on the first attempt.

2. PCA; Variance maximization & singular value decomposition

$X \in \mathbb{R}^{N \times d}$ X_c is centered. (each col has mean of 0) $Cov = C = \frac{1}{N-1} X_c^T X_c \in \mathbb{R}^{d \times d}$

a. First PC max variance of projection: $V \in \mathbb{R}^d$, Proj taken $z_n = V^T(x_n - \bar{x}) = \text{vector}^T(\text{Centroid data})$

I. Show var $z_1, \dots, z_n = V^T C V$ $z_n = \text{scalar}$ How far x_n projects onto V , $Var(z) = \frac{1}{N-1} \sum (z_n - \bar{z})^2$

$$\bar{z} = \frac{1}{N} \sum z_n = \frac{1}{N} \sum V^T(x_n - \bar{x}) = V^T \frac{1}{N} \sum (x_n - \bar{x}) = 0 \quad \text{so } \bar{z} = 0 \quad var(z) = \frac{1}{N-1} \sum z_n^2$$

$$= V^T \left(\frac{1}{N-1} \sum (x_n - \bar{x})^T (x_n - \bar{x}) \right) V = V^T C V$$

II. max $V^T C V$, $V^T V = 1$ $L(V, \lambda) = V^T C V - \lambda(V^T V - 1)$ $PCL = 2CV - 2\lambda V = 0$ $(CV = \lambda V)$

III. pick eigenvector with largest eigenvalue? C is $d \times d$ which ones? $V^T C V = V^T \lambda V = \lambda$ so eigenvector with largest eigenvalue max variance

b. SVD: $X_c = U \Sigma V^T$ $U \in \mathbb{R}^{N \times r}$, $\Sigma \in \mathbb{R}^{r \times r}$ diagonal, $V \in \mathbb{R}^{d \times r}$ $r = \min(N, d)$ $U^T U = V^T V = I$

I. $X_c^T = V \Sigma^T U^T = V \Sigma U^T$; $(V \Sigma U^T)(U \Sigma V^T) = V \Sigma U^T U \Sigma V^T = V \Sigma^2 V^T$

II. cols are eigenvectors of $C = X_c^T X_c / (N-1)$ if $\lambda_i = \sigma_i^2 / (N-1)$; $C = \frac{1}{N-1} X_c^T X_c = \frac{1}{N-1} V \Sigma^2 V^T$

$$Cv_i = \frac{1}{N-1} V \Sigma^2 V^T v_i = \frac{1}{N-1} V \Sigma^2 e_i = \frac{\sigma_i^2}{N-1} V e_i = \frac{\sigma_i^2}{N-1} v_i \quad \text{each col of } V \text{ is eigenvector of } C$$

III. SVD Preferred when d is large? C is $d \times d = 2 \times k \times 2 \times k$ very high mem consumption

Svd works directly on x so 801×20531 much smaller, when $N \ll d$ so 156×20531

Here are at most $N-1$ non-zero eigenvalues anyway so you're not losing anything by using SVD on smaller matrix